

Mini-Project Integration, Testing & Rehearsal

Welcome to Module 11 of your GENAI, NLP and Prompt Engineering course. Today marks a pivotal moment in your learning journey as you bring together all the components of your mini-project into a polished, presentation-ready system.

This session is designed to ensure your project runs smoothly, your explanations are clear and confident, and you're fully prepared for assessment day. You'll complete your setup, verify system functionality, rehearse your demo, and receive valuable peer feedback—all essential steps toward a successful final presentation.



MODULE 11



TODAY'S FOCUS



Task 1: Assemble the Mini-Project

What You Will Assemble

Bring together all the essential components of your project into one organized location. This includes your final Jupyter notebook or Python script containing your working code, the complete dataset or spreadsheet that powers your system, and a professional one-slide summary that clearly describes your project's purpose and functionality.

Think of this as creating your project's home base—everything should be accessible, labeled clearly, and ready to demonstrate at a moment's notice.

Expected Outcome

By the end of this task, all project components will be consolidated in a single, well-organized directory or workspace. You'll have verified that every file opens without errors, your code executes from start to finish, and your dataset loads correctly.

This organizational step eliminates last-minute technical surprises and gives you confidence that your system is deployment-ready for demonstration and assessment.

Task 2: Verify the System Output

Quality assurance is critical for any AI system. This task focuses on thorough testing to ensure your mini-project performs reliably and produces meaningful results under various conditions.

1

Test Input-Output Flow

Run your system with multiple test cases to confirm the input-to-output pipeline functions smoothly. Verify that data flows correctly through each processing stage and that the system handles different input types appropriately.

2

Assess Answer Quality

Evaluate whether your system's outputs are clear, accurate, and easy to understand. Check that responses directly address the input queries and that technical jargon is appropriately explained for your target audience.

3

Verify Citations & References

If your system provides sources or references, confirm they appear correctly and link to valid information. Proper attribution demonstrates academic integrity and helps users verify the reliability of generated content.

4

Check Fallback Behavior

Test how your system handles edge cases, unexpected inputs, or situations where it lacks sufficient information. Ensure appropriate fallback messages appear that guide users gracefully rather than producing errors or confusing outputs.

- ☐ **Outcome:** Your system behaves predictably and professionally across a range of scenarios, giving you confidence during demonstration and showcasing your attention to quality and user experience.

Task 3: Prepare Your 2-Minute Explanation

Clear communication is just as important as technical execution. You'll craft and practice a concise, compelling explanation of your project that demonstrates both your technical skills and your ability to communicate complex concepts effectively.

01

Define the Problem

Articulate what real-world challenge or need your project addresses. Help your audience understand why this problem matters and who benefits from your solution.

02

Describe Your Data

Explain what data sources power your system, how you obtained or prepared them, and why they're appropriate for solving your identified problem.

03

Showcase the Output

Demonstrate what your system produces—whether answers, visualizations, recommendations, or insights—and highlight what makes the output valuable or unique.

04

Acknowledge Limitations

Discuss at least one limitation of your system with honesty and insight. This demonstrates critical thinking and awareness of real-world constraints in AI development.

Practice makes perfect—rehearse your explanation multiple times until you can deliver it naturally within two minutes, without reading from notes. This preparation will help you feel confident and professional during your final presentation.

Task 4: Pair Rehearsal



Peer feedback is invaluable for identifying blind spots and refining your presentation. Working in pairs provides a safe space to practice, receive constructive criticism, and learn from your classmates' approaches and insights.

Rehearsal Process

- **Run your demo once** with your partner observing, showing the full input-output cycle of your system in action
- **Deliver your 2-minute explanation** while your partner takes notes on clarity, pacing, and completeness
- **Listen actively to your partner's presentation** and provide thoughtful, specific feedback on both technical and communication aspects
- **Exchange observations** about what worked well and what could be strengthened before the final assessment

❑ **Outcome:** You'll gain a clear understanding of your presentation's strengths and areas for improvement. This collaborative learning experience helps both partners refine their demos and builds confidence for assessment day.

Task 5: Finalize & Submit

Demo Files

Your complete, tested code notebook or script with all dependencies documented and functioning correctly. Ensure file naming is clear and professional.

Project Summary

A polished one-slide overview that captures your project's essence: problem, approach, data, output, and key insights in a visually appealing format.

Supporting Materials

Screenshots of your system in action, sample outputs, or visualizations that demonstrate functionality and showcase your best results.

Double-check that all components are properly labeled, your documentation is complete, and everything is ready for submission. Take a moment to review the submission requirements one final time to ensure nothing is missing.

What's Next: Module 12

You've done the hard work of building, testing, and refining your mini-project. Now it's time to showcase your achievement on **Assessment Day**, where you'll present your system and demonstrate the skills you've developed throughout this course.



ASSESSMENT DAY



MINI-PROJECT PRESENTATIONS

Congratulations on reaching this milestone! Your preparation today sets you up for success tomorrow. Trust in the work you've done and present with confidence.